Hello, I'm currently following a paper on the Color Glass Condensate. I'm trying to understand how the classical treatment of the saturated gluon fields is justified. First, they imply that the unintegrated gluon density is of the order of 1/\alpha_s, but I'm not sure where this estimate comes from. How is it justified?

This is a very standard and important question in the CGC framework, and it's good you're pausing on it — the whole classical-field picture hinges on this estimate.

Let me go step by step and keep the logic explicit.

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1. What quantity is being estimated?

In CGC one often states that, for transverse momenta below the saturation scale,

$$k_\perp \lesssim Q_s,$$

the **occupation number** (or unintegrated gluon density per phase–space cell)

$$f(k_\perp) \sim \frac{1}{\alpha_s}.$$

This is *not* the integrated PDF $xG(x, Q^2)$, but rather the number of gluons per unit transverse phase space and rapidity.

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2. Physical meaning: large occupation number

The key idea is:

- Classical fields are justified when the occupation number is large:

$$f \gg 1.$$

- In non-Abelian gauge theory, the natural scale separating "large" and "small" is $1/\alpha_s$, because interactions involve powers of $g = \sqrt{4\pi\alpha_s}$.

So the statement $f \sim 1/\alpha_s$ is saying:

the gluon fields are as large as they can be without becoming nonperturbative in the coupling.

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3. Heuristic derivation from field strength

Start with the Yang–Mills action:

$$S \sim \frac{1}{g^2} \int d^4x \, F^a_{\mu\nu} F^{\mu\nu a}.$$

If the classical field dominates, the action should be large compared to $\hbar$ (set to 1):

$$S \gg 1.$$

This suggests that the typical field strength scales as

$$A_\mu \sim \frac{1}{g}.$$

Then the chromo-electric and magnetic fields scale as

$$F_{\mu\nu} \sim \partial A + gA^2 \sim \frac{1}{g}.$$

So **large classical fields** naturally have amplitude $A \sim 1/g$.

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4. From field amplitude to occupation number

Now connect this to gluon occupation numbers.

For a mode with momentum k, the energy density behaves like

$$\epsilon \sim k^2 A^2.$$

The number density of gluons in that mode is roughly

$$n_k \sim \frac{\epsilon}{k} \sim kA^2.$$

Using $A \sim 1/g$,

$$n_k \sim \frac{k}{g^2}.$$

The occupation number per phase-space cell (unintegrated gluon distribution) is

$$f(k) \sim \frac{n_k}{k^3} \sim \frac{1}{g^2} \sim \frac{1}{\alpha_s}.$$

This is the parametric estimate quoted in CGC papers.

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5. Saturation argument (more intuitive)

There is also a **dynamical** argument based on saturation:

- At small x, gluon splitting increases the density.
- Recombination becomes important when

$$\sigma_{\text{recomb}} \times \text{density} \sim 1.$$

- The recombination cross section is perturbative:

$$\sigma \sim \frac{\alpha_s}{k_\perp^2}.$$

- Setting the density $n \sim f\, k_\perp^2$ and requiring balance gives

$$f \sim \frac{1}{\alpha_s}.$$

Thus saturation occurs precisely when the occupation number reaches $1/\alpha_s$.

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6. Why this justifies a classical treatment

- Even though $\alpha_s(Q_s) \ll 1$, the fields are strong:

$$gA \sim \mathcal{O}(1).$$

- Loop corrections are suppressed by extra powers of α_s.
- Quantum evolution in x is encoded in the **JIMWLK/BK equations**, while the fields themselves are classical solutions of Yang–Mills equations with stochastic sources.

Hence:

> **weak coupling but strong classical fields**.

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7. Summary in one sentence

The estimate $f \sim 1/\alpha_s$ comes from the fact that saturation occurs when gluon recombination balances splitting, which happens at occupation numbers of order $1/\alpha_s$; equivalently, classical Yang–Mills fields with amplitude $A \sim 1/g$ naturally correspond to gluon densities of order $1/\alpha_s$.

If you'd like, next we can:

- connect this estimate explicitly to the McLerran–Venugopalan model, or
- derive it using Wilson lines and the dipole amplitude, or
- clarify the difference between $xG(x, Q^2)$ and the unintegrated distribution in CGC.